3 Preperation with Mechanism

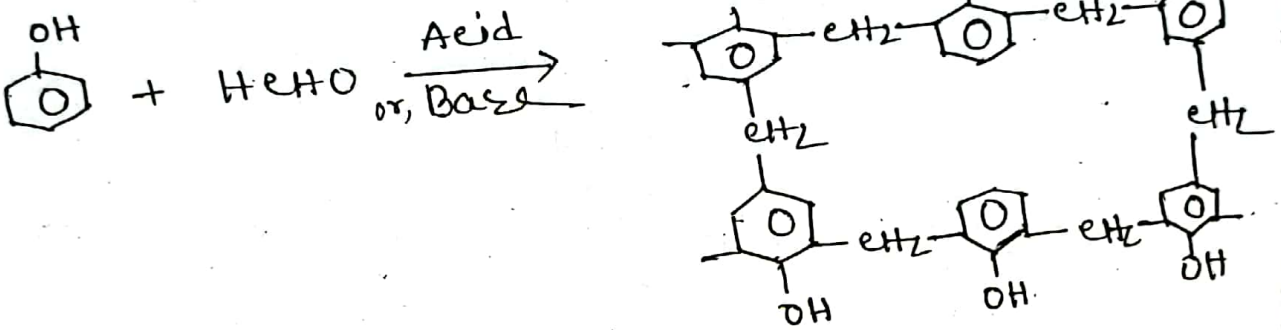
5th

resin

TAPOSY

Phenol – Formaldehyde resin

Phenol-Formaldehyde resin is also called phenolic resin. Phenol-formaldehyde resins are synthetic polymers obtained by the reaction of phenol or substituted phenol with formaldehyde.



Bakelite

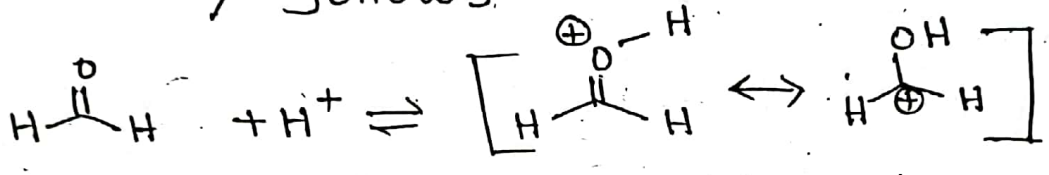
5 marks

18 Mechanism :Novalacs

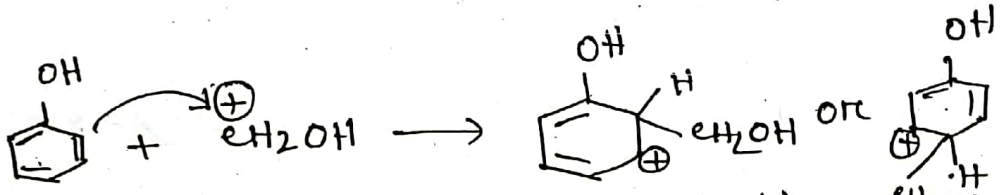
The reaction between phenol and HCHO is catalysed either by acid or base. The process involves the formation of methylene bridge in ortho ; para or both ortho-para position. This

results in the formation of linear polymers Novolac then the cross linked polymers Bakelite.

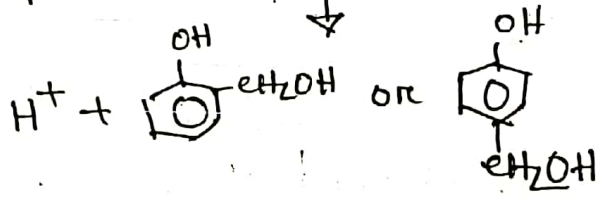
Acid catalysed polycondensation proceed as follows—

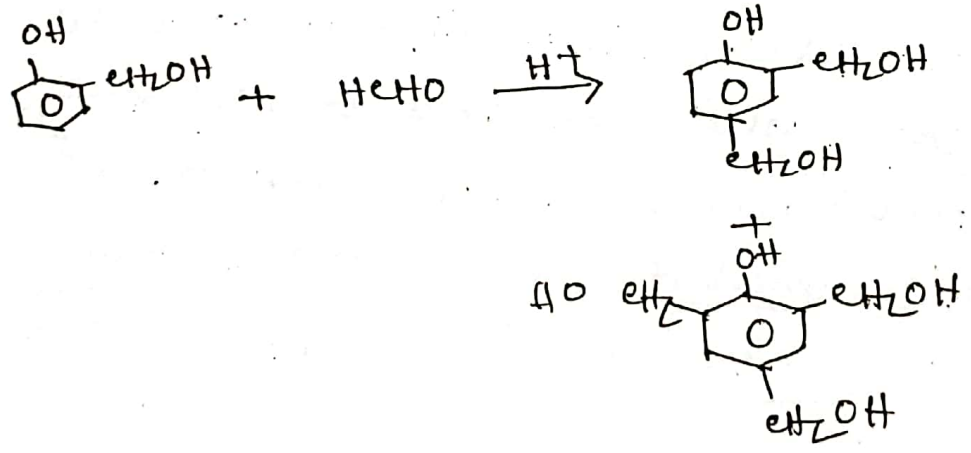
1. 

Electrophilic addition

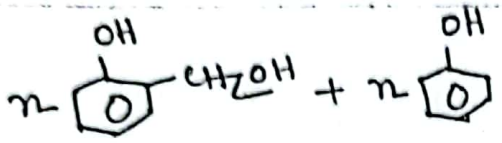
2. 

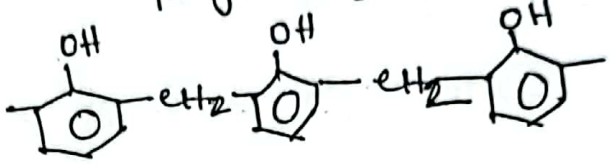
Electrophilic substitution



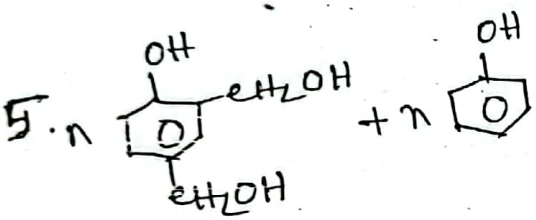
3. 

Novalac

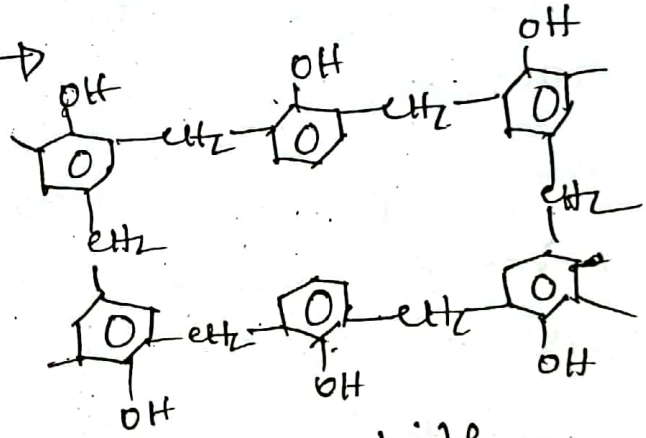
4. 



16/ ortho → Novolac
Linear polymer

5. 

cross link



ortho para → Bakelite

P-F = Phenol Formaldehyde

properties :

* P-F resins having low degree of polymerization are soft. They possess excellent adhesive properties.

* P-F resins having high degree of polymerization are hard, rigid and infusible. They are resistant to reducing agents, salts and many organic solvents. They can withstand very high temp. They act as excellent electrical insulators.

molding
electronics
Automobils, fabrics

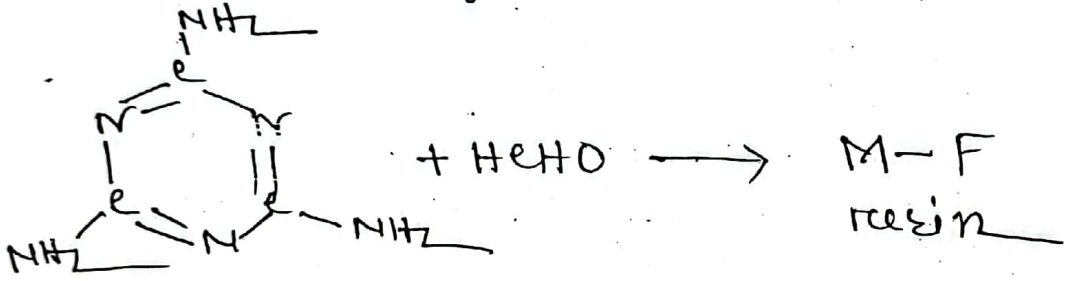
uses :

* They are used for making moulded articles such as radio and tv parts, fountain pen, photograph records etc.
* They are used formaking decorative laminates, wall covering etc.
* They are used for making electrical goods such as switch, plugs.
* They are used a bonding glue fore laminated wooden planeks and in varnish.
* They are used fore impregnating fabrics wood and paper.
* Sulfonated p-F resin is used as ion-exchange resin.

Melamine-formaldehyde resin

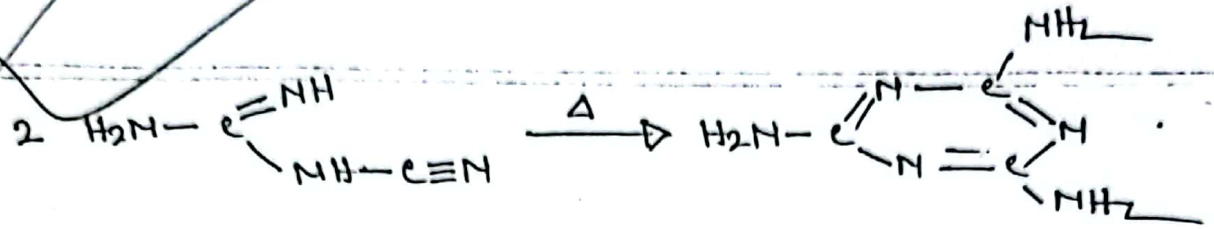
Melamine-formaldehyde resin is prepared by the polycondensation of melamine and formaldehyde.

preparation:



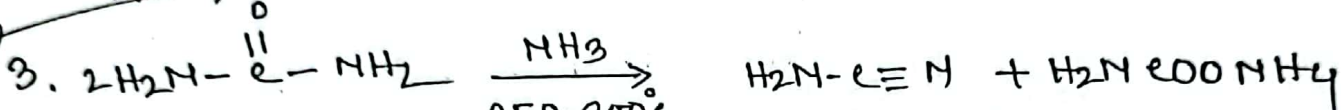
Mechanism: from calcium carbide

1. $CaC_2 + N_2 \longrightarrow$ CaN·CN (calcium cyanamide) $\xrightarrow{H_2SO_4}$ $H_2N-C\equiv N$ (cyanamide) $\xrightarrow{100^\circ C}$ $H_2N-C(=NH)-NH-C\equiv N$ (Dicyanamide)

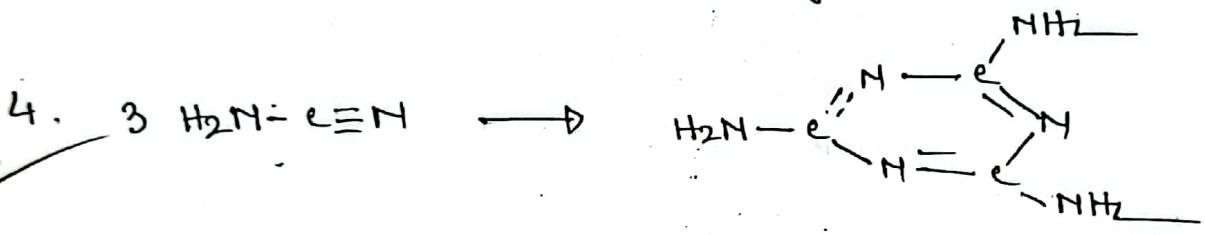
2. 

Melamine

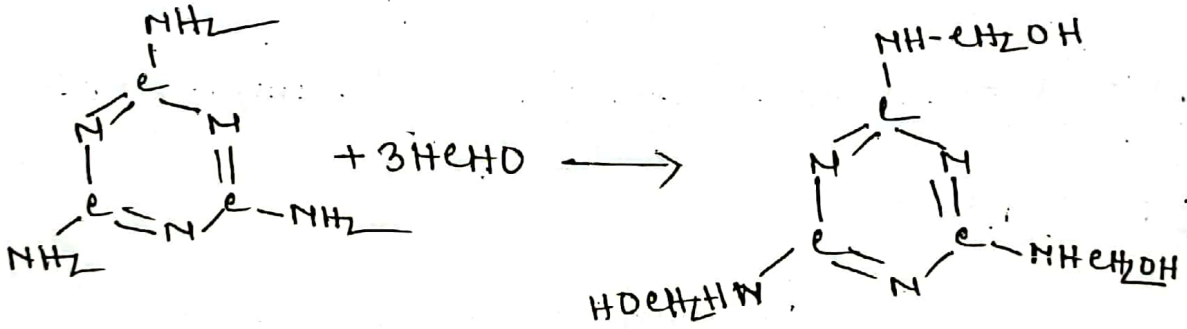
From urea

3. 

Cyanamide

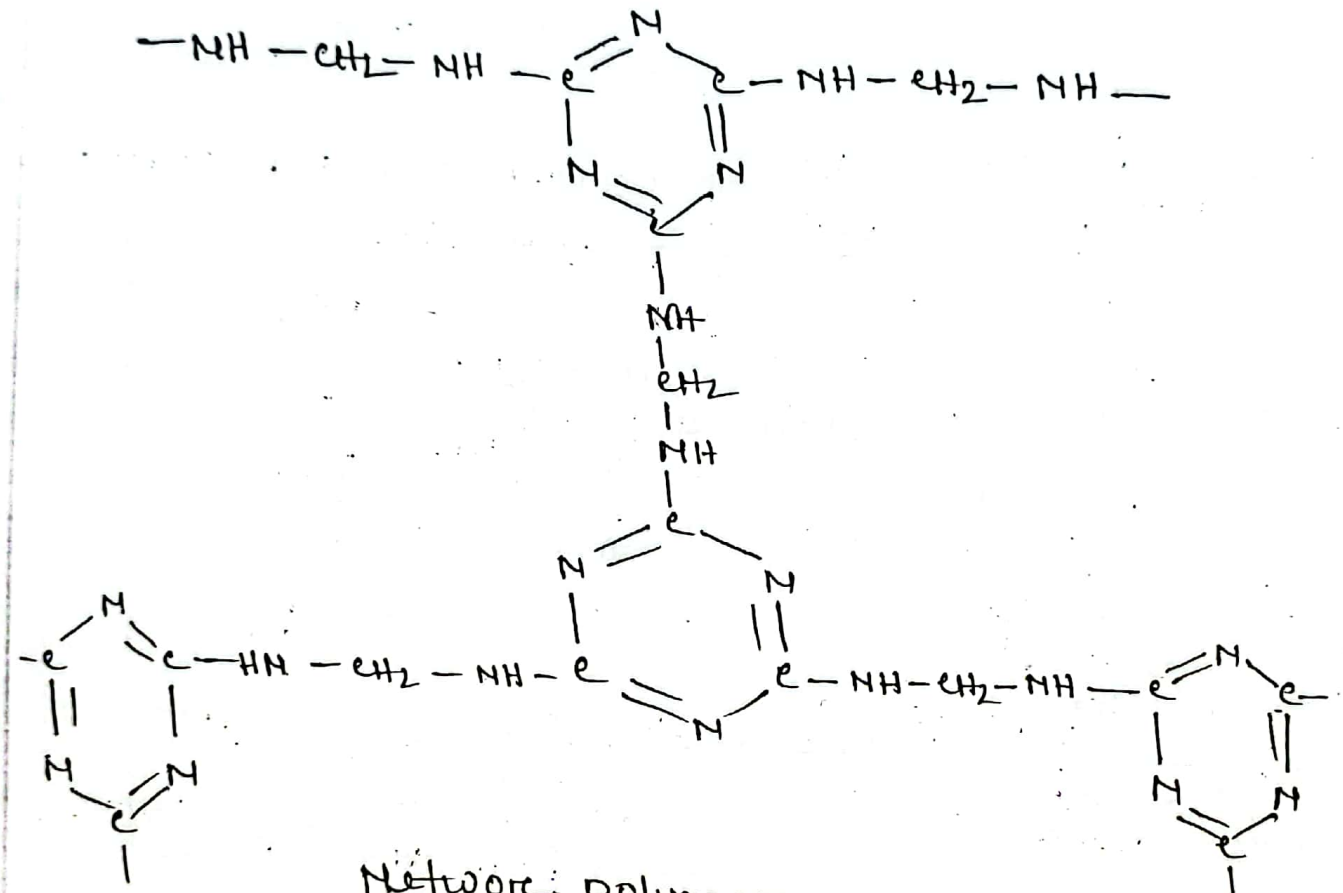
4. 

Melamine

5. 

Trimethylo melamine

If excess melamine is used then we will get linear polymer, But if we used excess amount of formaldehyde then we will get three dimentional or network polymer.



Network polymers

M-F

Properties :

M-F
* Moisture resistant
* More Harder and more stronger

U-F
* Rigid, insoluble and infusible
* High tensile strength
* High surface hardness
* High heat distortion temperature
* Low H_2O absorption
* Low refraction index
* It is nontransparent thermosetting plastic.
* Elongation at break and volume resistance.

Melamine-formaldehyde (M-F) Resins

Uses :

* They are used for decorative laminates
* Modification of textile fibre
* It is a thermosetting plastic used in high pressure
* Thermal and fire retardancy
* Sulfonated M-F used as cement admixture
* It is used as super plasticizer

from nature

Urea-formaldehyde resin

The U-F resins are obtained by the polycondensation of urea and formaldehyde.

preparation:

$$H_2N-\overset{\overset{O}{\|}}{C}-NH_2 + HCHO \longrightarrow \text{U-F resin}$$

$$H-[HN-\overset{\overset{O}{\|}}{C}-NH-CH_2-]_n-OH$$

Mechanism:

1. $O=C(NH_2)_2 + HCHO \longrightarrow O=C(NH-CH_2OH)(NH_2)$

$O=C(NH-CH_2OH)(NH_2) + HCHO \longrightarrow O=C(NH-CH_2OH)_2$

2. $O=C(NH-CH_2OH)_2 + H_2N-\overset{\overset{O}{\|}}{C}-NH_2 \longrightarrow$

$O=C(NH-CH_2OH)(NH-CH_2-NH-\overset{\overset{O}{\|}}{C}-NH_2) + H_2O$

3. $n\ H_2N-\overset{\overset{O}{\|}}{C}-NH_2 + n\,HCHO \longrightarrow$

$$H-\left[HN-CO-NH-CH_2\right]_n-OH$$

If excess formaldehyde is used then the H atom of the imide group can easily replaced.

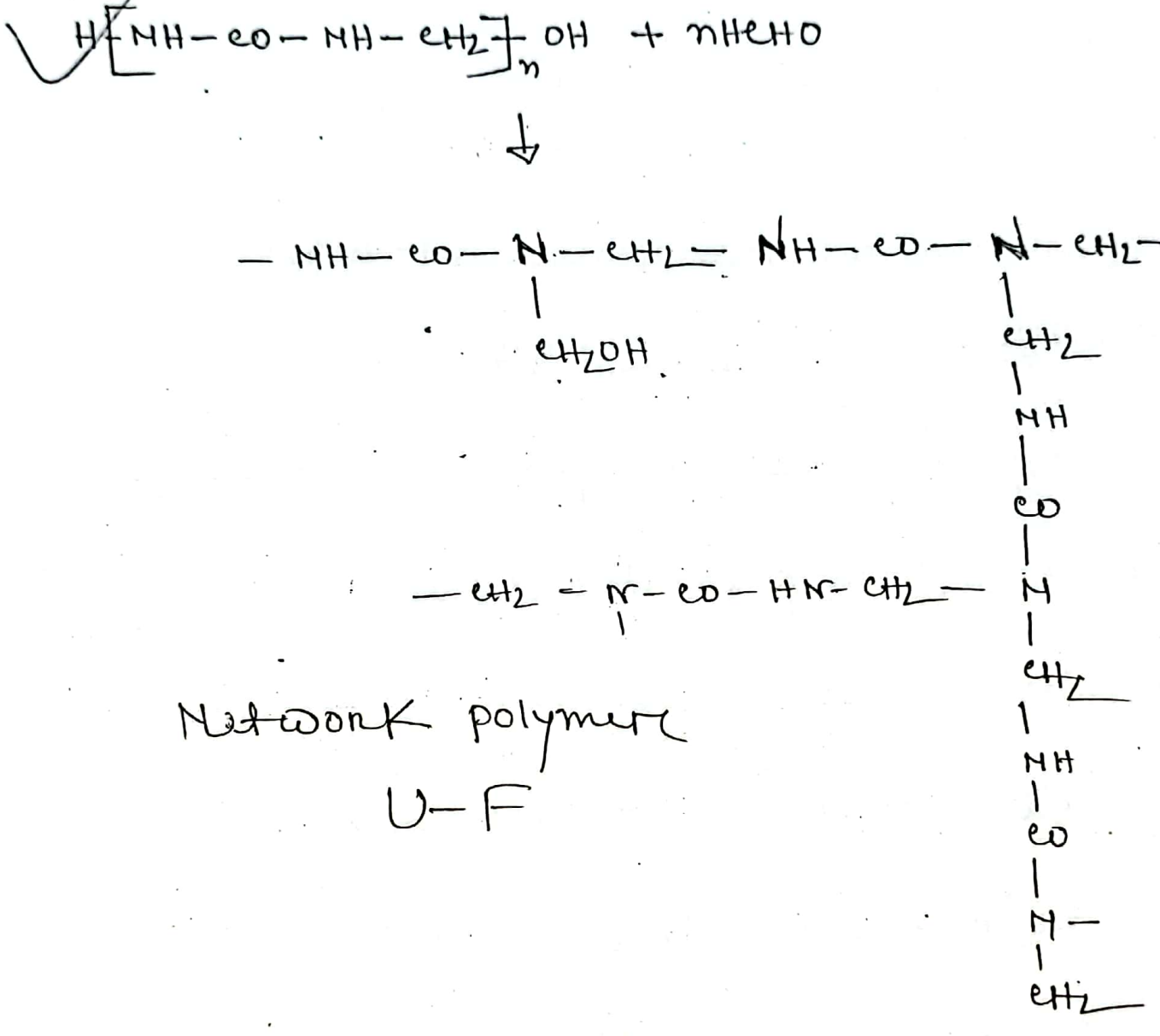
$$-HN-CO-NH-CH_2- + HCHO$$

$$\downarrow$$

$$-HN-CO-\underset{\underset{CH_2OH}{|}}{N}-CH_2-$$

The final composition depends upon the ratio of urea and formaldehyde.

If the linear polymere is heated with excess formaldehyde then the network polymere is produced



Network polymer

U-F

properties :

* It is a theremosetting plastic used in high pressure
* It is a non-transparent resin
* Rigid, insoluble, infusible
* High tensile strength
* High surface hardness
* High heat distortion temp
* Low water absorption
* Refraction index . 1.54
* Elongation at break and volume resistance nature.

Uses :

* It is used as bonding glue for laminated wooden planeks.
* It is used in agriculture as a control release source of N-fertilizer
* It is used for making moulded article and adhesives
* It is used for making decorative laminates
* It is used in making electrical material casing.

Rubber:

Rubber is an elastomere. It is an elastic hydrocarbon polymere. Rubber can be natural or manmade.

properties:

* Rubber is flexible
* Elastic.
* Insulator of electricity
* poor conductor of heat.
* Rubber is water proof
* Not transparent
* resistant to heat and chemicals

F-15

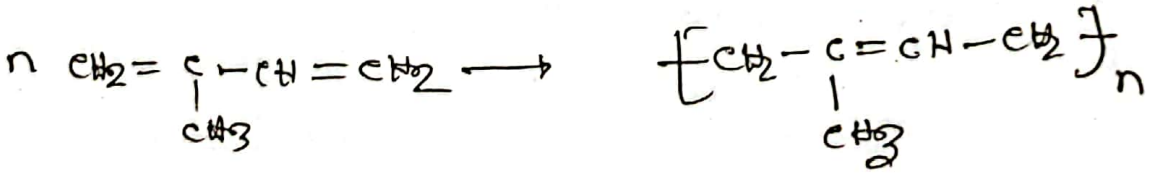
18

Natural Rubber:

Natural rubber is also known as 'india rubber as initially prepared. Natural rubber consists of polymer of the organic compound isoprene with minor impurities of other organic compounds plus water.

uses:

* Elastomere
* It is used as adhesives in cement
* Insulating and friction tapes
* Insulating blankets and footwear
* Tire
* Hoses / ~~Houses~~ Hosey.
* wheel rings



Isoprene
[2-methyl-1,3-butadiene]

natural rubber
poly-isoprene.

Synthetic rubber:

Synthetic rubber consists of polymer of the petroleum based precursor called monomer. The ~~first or~~ most common synthetic rubber is (SBR) styrene-butadiene rubber

uses:

* ~~*~~ car tire
* flexible rubber toys
* paints
* shoe shole
* Rubber gloves
* tubes and Hoses

chloroprene undergoing trans-polymerization to produce synthetic rubber. polychloroprene.



chloroprene

~~Polyesters, Polyamide~~

nature

Natural rubber	Synthetic rubbere
* It is found in the sap of some plants especially rubber tree.	* It is Artificially made rubbere which acts as elastomere
* It is weaker	* It is Strongere
* Thermal stability is low	* Thermal stability is high
* It is also known as india rubbere	* It is not so
* It is naturally occeured	* It is man made

F-18

Polyvinyl chloride

Polymer of vinyl chloride

$$n.CH_2=CH-Cl \xrightarrow[\text{Polymerization}]{} \left[CH_2-\underset{Cl}{\underset{|}{CH}}\right]_n$$

Properties :

* It has high density
* rigid, insoluble.
* It is colourless

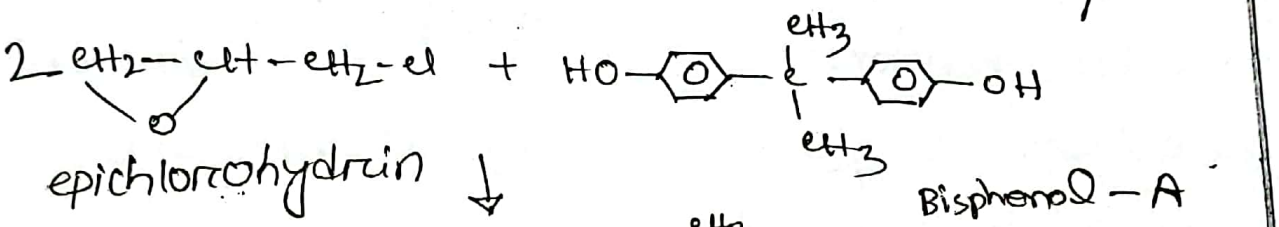
Uses :

* It is used in cable insulation
* Fibre manufacturing
* It is used as carpet of house floor.
* making raincourt
* pipe
* pakaging, toys
* laminated materials
* plastic syringes
* equipment parts.

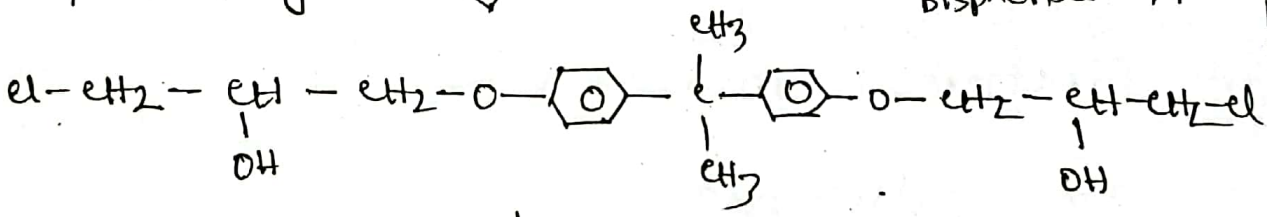
f-16, 13, 18

Epoxy polymer.

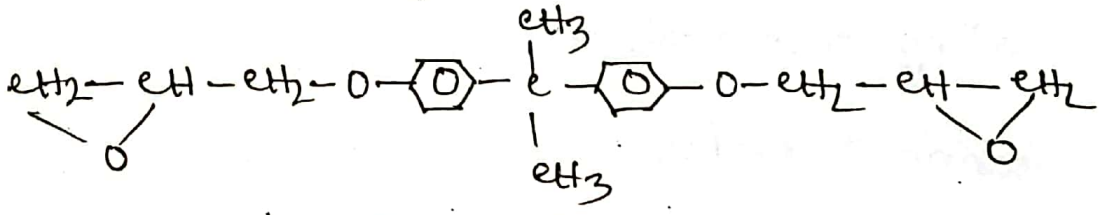
Epoxy polymer is basically polyether resin. It is prepared from Epichlorohydrin and bis-phenol-A. The reaction is carried out on excess epichlorohydrin



epichlorohydrin ↓ Bisphenol-A



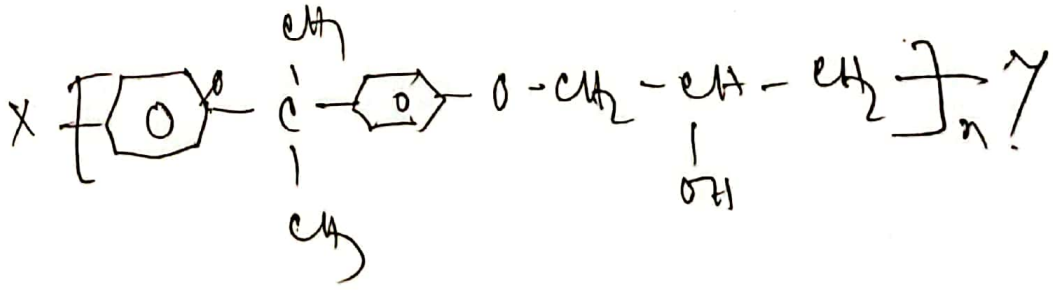
↓ NaOH (-NaCl)



Epoxy polymer

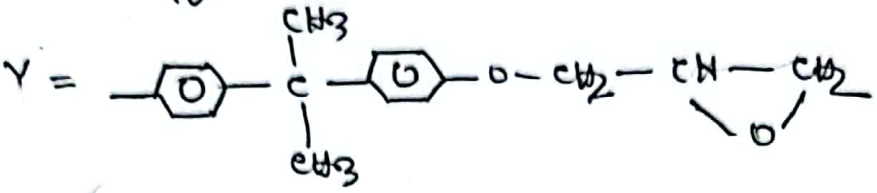
+ Bisphenol-A (excess.)

↓



where,

$X = -CH_2-CH(O)CH_2-CH_2-O-$ and



E-16

properties :

* good Electrical insulator
* Better thermal stability
* chemical resistance except alkali.
* heat resistance.

Uses :

* Electrical insulation
* good adhesive
* coating compounds
* sealants varnishes and paints.

nature

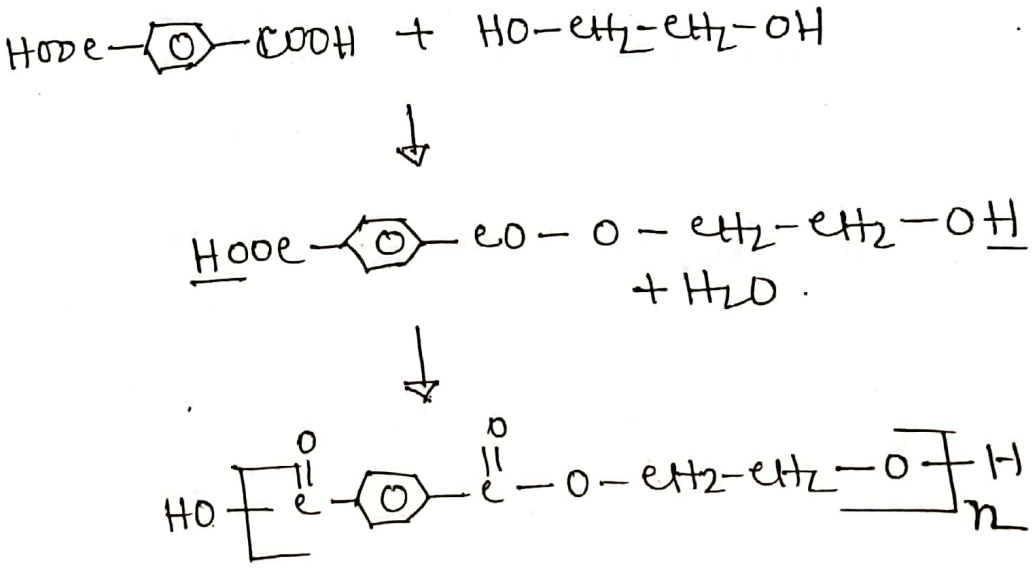
12

polyesters

polyesters is obtained either by self condensation of a hydroxy carboxylic acid or by the reaction of a dicarboxylic acid and a glycol.

Example:

Terylene is obtained by the reaction of 1,4-dicarboxylic acid and ethane-1,2 diol.



f.16 Uses :

* It is used to make fibre
* It is used to make films which are used to make sheets.
* It has good mechanical strength

polymere resin 18

polymere resin is a cleare liquid plastic product that hardens to create thick, durable and glossy coating

polymere resin is a cleare liquid plastic product that hardens to create thick, durable, glossy coating

polymere resin is a cleare liquid plastic product that hardens to create thick, durable, glossy coating,

polymere resin is a cleare liquid plastic product that hardens to create ~~du~~ thick, durable and glossy coating